AD7031

Integer Programming In Machine Learning

2026 Fall

KNDU

AI Applications







ChatGPT

Gemini

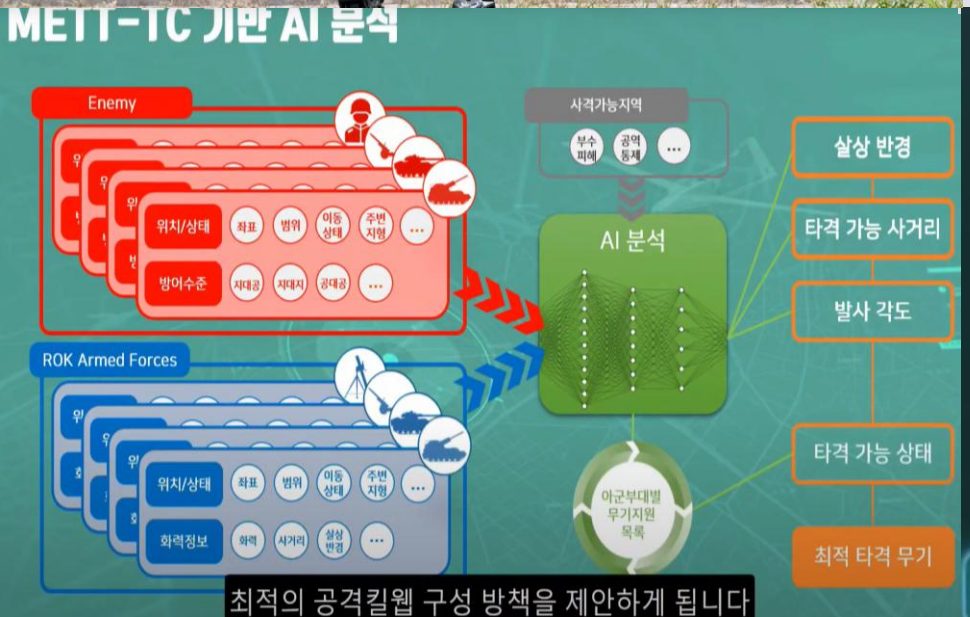
perplexity

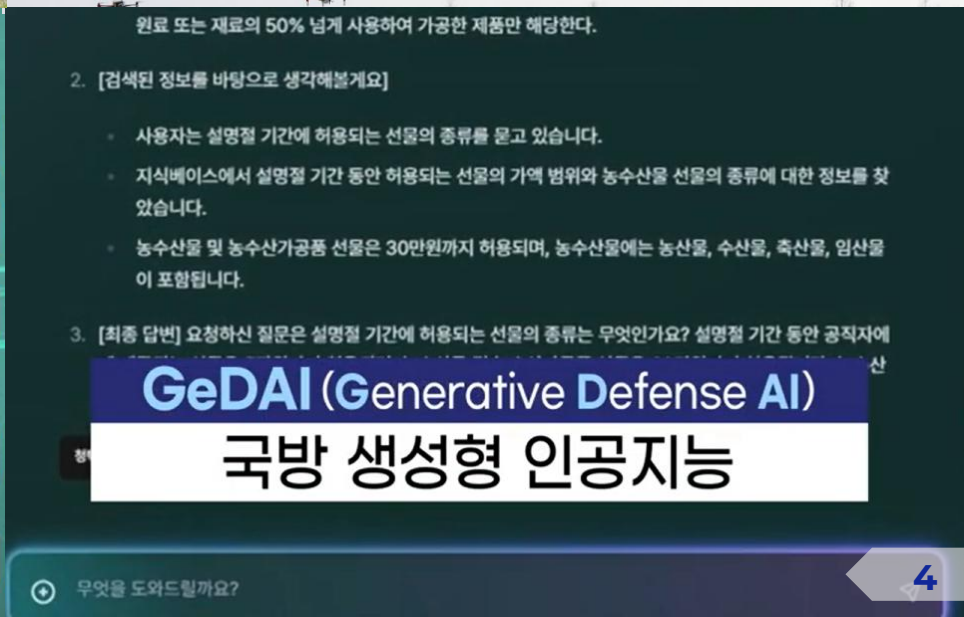
Claude

AI in Defense: Use Cases

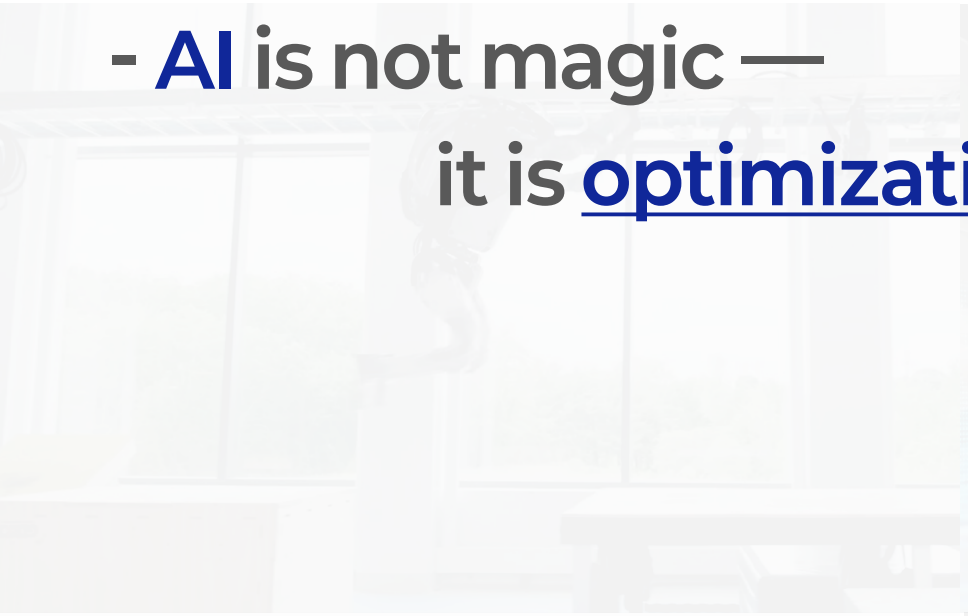




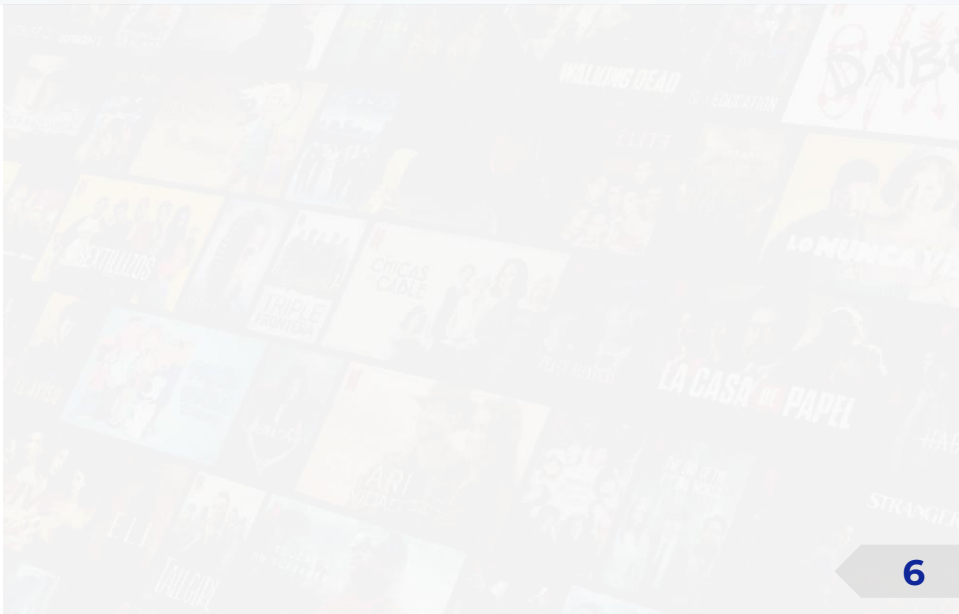




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it is optimization under uncertainty







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- ML is just a function approximation :

$$f_{\theta}(\boldsymbol{x}) \approx f^{true}(\boldsymbol{x})$$

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One problem, three languages

	OR	ML	Control
Decision	$\boldsymbol{x}$	$\boldsymbol{\theta}$	$\boldsymbol{u}$
Uncertain data	$\tilde{\xi}$	$[\tilde{x}; \tilde{y}]$	State $\tilde{x}$ by noise $\tilde{w}$

Optimization under Uncertainty: ML

Random vector $\tilde{\xi} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \end{bmatrix}$ follows distribution $\mathbb{P}$ (i.e., $\tilde{\xi} \sim \mathbb{P}$), then the best model parameter $\theta^\star$ is

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Many **loss (objective) functions** $\ell(\cdot)$ are used in ML

- Regression : predict continuous $\boldsymbol{y} \in \mathbb{R}^n$
- Classification : predict binary $\boldsymbol{y} \in \{-1, +1\}$

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Squared Loss$(\cdot) = \|\cdot\|_2^2 = \|y - f_\theta(x)\|_2^2$

e.g., when n=2

$$y_1 = \begin{bmatrix} 100 \\ 10 \end{bmatrix}, f_\theta(x_1) = \begin{bmatrix} 105 \\ 9 \end{bmatrix}$$

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$$\Rightarrow \left\| \begin{bmatrix} 100 \\ 10 \end{bmatrix} - \begin{bmatrix} 105 \\ 9 \end{bmatrix} \right\|_2^2 = \left\| \begin{bmatrix} -5 \\ 1 \end{bmatrix} \right\|_2^2 = (-5)^2 + 1^2 = 26$$

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$$\Rightarrow \|10 - 11\|_2^2 = \|-1\|_2^2 = (-1)^2 = 1$$

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$$\text{0–1 Loss}(\cdot) = \begin{cases} 0, & \text{if } \operatorname{sign}\left(f_\theta(x)\right) = \operatorname{sign}(y) \\ 1, & \text{if } \operatorname{sign}\left(f_\theta(x)\right) \neq \operatorname{sign}(y) \end{cases}$$

e.g., $y_1 = -1, f_\theta(x_1) = -1000.5$

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$y_2 = +1, f_\theta(x_2) = -1$

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e.g., $y_1 = -1, f_\theta(x_1) = -1000.5 \Rightarrow$ Correct! 0-loss

$y_2 = +1, f_\theta(x_2) = -1 \Rightarrow$ Wrong! 1-loss

Optimization under Uncertainty: ML

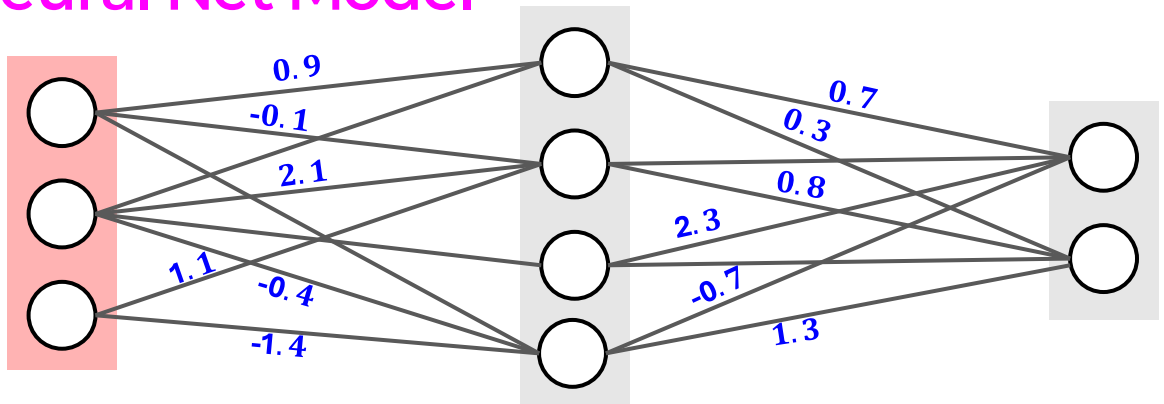
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Linear Model

$$f_\theta(\tilde{x}) = \theta^\top x = \sum_i x_i \theta_i$$

Neural Net Model



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ML $\subset$ optimization under uncertainty

- OR :

$$\min_{x} \ \mathbb{E}_{\mathbb{P}}\left[f\left(x, \tilde{\xi}\right)\right]$$
$$\text{s.t} \ \ \text{Prob}\left[g\left(x, \tilde{\xi}\right) \le 0\right] \ge \alpha$$

- Control

$$\min_{u_1, u_2, \ldots, u_{T-1}} \ \mathbb{E}_{\mathbb{P}}\left[\sum_{t=1}^{T-1} c_t\left(\tilde{x}_t, u_t, \tilde{\xi}_t\right) + c_T\left(\tilde{x}_T\right)\right]$$
$$\text{s.t} \ \ \tilde{x}_{t+1} = f_t\left(\tilde{x}_t, u_t, \tilde{\xi}_t\right) \ \forall t = 1, \ldots, T-1$$

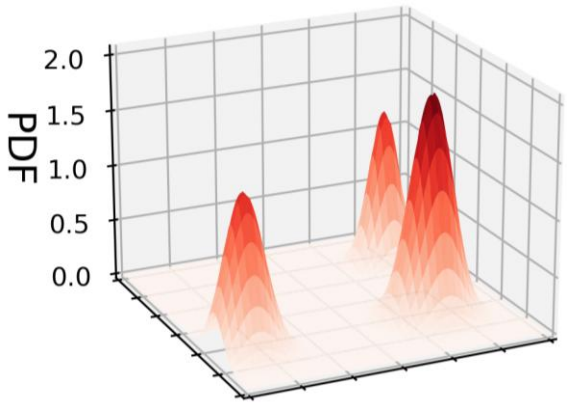
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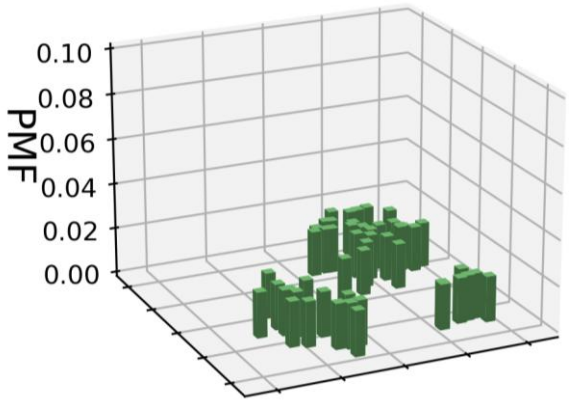
$$\theta^\star \in \arg\min_{\theta} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\tilde{y}, f_\theta(\tilde{x})\right)\right]$$

$\mathbb{P}$ unknown $\Longrightarrow$ $\theta^\star$ not available

Only N samples $= \{\xi_1, \xi_2, \ldots, \xi_N\}$ available: $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$



True distribution $\mathbb{P}$



Empirical distribution $\widehat{\mathbb{P}}_N$

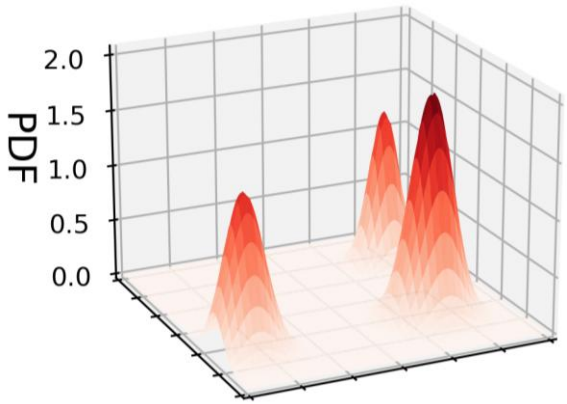
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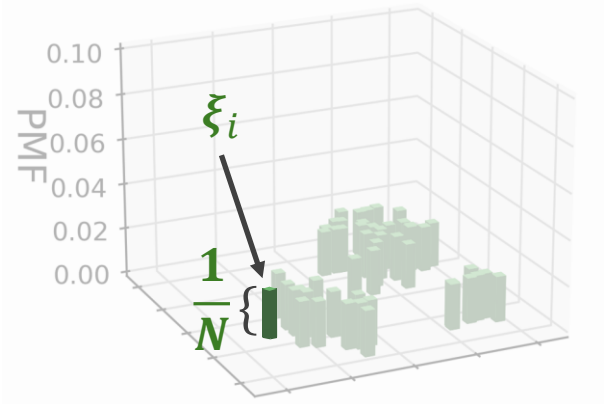
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True distribution $\mathbb{P}$



Empirical distribution $\hat{\mathbb{P}}_N$

Optimization under Uncertainty: ML

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$$\theta^\star \in \arg\min_{\theta} \mathbb{E}_{\mathbb{P}}\left[\ell\left(y, f_\theta(x)\right)\right]$$

Data-Driven Optimization

Assume $\xi \sim$empirical distribution $\widehat{\mathbb{P}}_N$, then the best empirical parameter $\widehat{\theta}_N$ is

$$\widehat{\theta}_N \in \arg\min_{\theta} \mathbb{E}_{\widehat{\mathbb{P}}_N}\left[\ell\left(y, f_\theta(x)\right)\right]$$

$$= \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f_\theta(x_i))$$

Optimization under Uncertainty: ML

True Problem

$$\boldsymbol{\theta}^\star \in \arg\min_{\boldsymbol{\theta}} \mathbb{E}_{\mathbb{P}}\left[\ell\left(\boldsymbol{y}, f_{\boldsymbol{\theta}}(\boldsymbol{x})\right)\right]$$

$$\neq$$

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Optimization under Uncertainty: ML

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Data-Driven Problem

If $\hat{\mathbb{P}}_N \approx \mathbb{P}$, then $\hat{\theta}_N$ is a good approximation to $\theta^\star$

- In ML , $\hat{\theta}_N$ is called Empirical Risk Minimizer (ERM)
- In OR , it is called Sample Average Approximation (SAA)

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$

$$\widehat{\theta}_N \in \operatorname*{argmin}_{\theta} \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f_\theta(x_i))$$

Goal : Predict binary output given input data

- Output $y_i \in \{-1, +1\}$

  e.g., -1 =Male, $+1$ =Female

- Input $x_i \in \mathbb{R}^{d+1}$

  e.g., x_i = [name, height, … , wieght, 1] why add 1? ⇒ bias term

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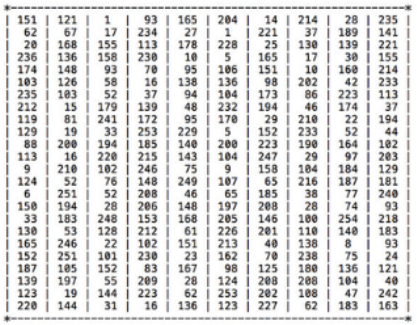
- Input $\boldsymbol{x}_i \in \mathbb{R}^{d+1}$

  e.g., $\boldsymbol{x}_i = [$Pixels of image i, 1$]$

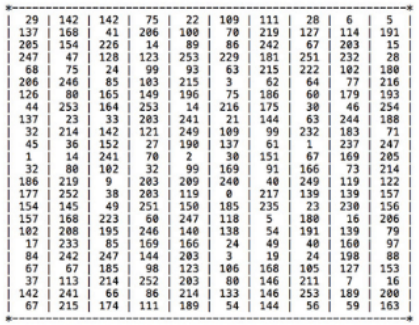
Q. what is d in this case?







151	121	1	93	165	204	14	214	28	235
62	67	17	234	27	1	221	37	189	141
20	168	155	113	178	228	25	130	139	221
236	136	158	230	10	5	165	17	30	155
174	148	93	70	95	106	151	10	160	214
103	126	58	16	138	136	98	202	42	233
235	103	52	37	94	104	173	86	223	113
212	15	179	139	48	232	194	46	174	37
119	81	241	172	95	170	29	210	22	194
129	19	33	253	229	5	152	233	52	44
88	200	194	185	140	200	223	190	164	102
113	16	220	215	143	104	247	29	97	203
9	210	102	246	75	9	158	104	184	129
124	52	76	148	249	107	65	216	187	181
6	251	52	208	46	65	185	38	77	240
150	194	28	206	148	197	208	28	74	93
33	183	248	153	168	205	146	100	254	218
130	53	128	212	61	226	201	110	140	183
165	246	22	102	151	213	40	138	8	93
152	251	101	230	23	162	70	238	75	24
187	105	152	83	167	98	125	180	136	121
139	197	55	209	28	124	208	200	104	40
123	19	144	223	62	253	202	108	47	242
220	144	31	16	136	123	227	62	183	163



29	142	142	75	22	109	111	28	6	5
137	168	41	206	100	70	219	127	114	191
205	154	226	14	89	86	242	67	203	15
247	47	128	123	253	229	181	251	232	28
68	75	24	99	93	63	215	222	102	180
206	246	85	103	215	3	62	64	77	216
126	80	165	149	196	75	186	60	179	193
44	253	164	253	14	216	175	30	46	254
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32	214	142	121	249	109	99	232	183	71
45	36	152	27	190	137	61	1	237	247
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45	36	152	27	190	137	61	1	237	247
1	14	241	70	2	30	151	67	169	205
32	80	102	32	99	169	91	166	73	214
186	219	9	203	209	240	40	249	119	122
177	252	38	203	119	0	217	139	139	157
154	145	49	251	150	185	235	23	230	156
157	168	223	60	247	118	5	180	16	206
102	208	195	246	140	138	54	191	139	79
17	233	85	169	166	24	49	40	160	97
84	242	247	144	203	3	19	24	198	88
67	67	185	98	123	106	168	105	127	153
37	113	214	252	203	80	146	211	7	16
142	241	66	86	214	133	146	253	189	200
67	215	174	111	189	54	144	56	59	163

Q. what is d in this case?

A. $d = 10^2 = 10$ pixles$\times 10$ pixles

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} \boldsymbol{x}_i \\ y_i \end{bmatrix}\right)$

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \ell(y_i, f_{\boldsymbol{\theta}}(\boldsymbol{x}_i))$$

Choose linear model : $f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\theta}^{\top} \boldsymbol{x} = x_1\theta_1 + \cdots + x_d\theta_d + 1\theta_{d+1}$

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} \boldsymbol{x}_i \\ y_i \end{bmatrix}\right)$

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Binary Classification

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Choose linear model : $f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\theta}^\top \boldsymbol{x} = x_1\theta_1 + \cdots + x_d\theta_d + 1\theta_{d+1}$

Choose 0-1 loss : $\ell(y_i, \boldsymbol{\theta}^\top \boldsymbol{x}_i) = \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$

Why $\mathbb{I}[\boldsymbol{y}(\boldsymbol{\theta}^\top \boldsymbol{x}) \leq 0]$?

$$\text{0–1 loss}(\cdot) = \begin{cases} 1, & \text{if } \operatorname{sign}(\boldsymbol{\theta}^\top \boldsymbol{x}) \neq \operatorname{sign}(\boldsymbol{y}) \\ 0, & \text{if } \operatorname{sign}(\boldsymbol{\theta}^\top \boldsymbol{x}) = \operatorname{sign}(\boldsymbol{y}) \end{cases} = \mathbb{I}[\boldsymbol{y}(\boldsymbol{\theta}^\top \boldsymbol{x}) \leq 0]$$

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[y_i(\theta^\top x_i) \leq 0]$$

Choose linear model : $f_\theta(x) = \theta^\top x = x_1\theta_1 + \cdots + x_d\theta_d + 1\theta_{d+1}$

Choose 0-1 loss : $\ell(y_i, \theta^\top x_i) = \mathbb{I}[y_i(\theta^\top x_i) \leq 0]$

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} \boldsymbol{x}_i \\ y_i \end{bmatrix}\right)$

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Choose 0-1 loss : $\ell(y_i, \boldsymbol{\theta}^\top \boldsymbol{x}_i) = \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$

- $y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0 \Rightarrow$ misclassification

  e.g., $y_i = -1$ and $\boldsymbol{\theta}^\top \boldsymbol{x}_i = 2.4 > 0 \Rightarrow \mathbb{I}[-1\times(2.4) \leq 0] = 1$

  $y_i = +1$ and $\boldsymbol{\theta}^\top \boldsymbol{x}_i = -1.7 < 0 \Rightarrow \mathbb{I}[+1\times(-1.7) \leq 0] = 1$

- $y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) > 0 \Rightarrow$ classification

  e.g., $y_i = -1$ and $\boldsymbol{\theta}^\top \boldsymbol{x}_i = -1.3 > 0 \Rightarrow \mathbb{I}[-1\times(-1.3) \leq 0] = 0$

  $y_i = +1$ and $\boldsymbol{\theta}^\top \boldsymbol{x}_i = +3.1 < 0 \Rightarrow \mathbb{I}[+1\times(+3.1) \leq 0] = 0$

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$$

Choose linear model : $f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \boldsymbol{\theta}^\top \boldsymbol{x} = x_1\theta_1 + \cdots + x_d\theta_d + 1\theta_{d+1}$

Choose 0-1 loss : $\ell(y_i, \boldsymbol{\theta}^\top \boldsymbol{x}_i) = \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$

	$\boldsymbol{\theta}^\top \boldsymbol{x}_i < 0$	$\boldsymbol{\theta}^\top \boldsymbol{x}_i > 0$
$y_i = -1$	Correct : $\ell(\cdot) = 0$	Misclassified : $\ell(\cdot) = 1$
$y_i = +1$	Misclassified : $\ell(\cdot) = 1$	Correct : $\ell(\cdot) = 0$

Goal : Among N samples, find the best $\boldsymbol{\theta}$ that matches

$$\text{sign}(\boldsymbol{\theta}^\top \boldsymbol{x}_i) = \text{sign}(y_i)$$

as many as possible

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} \boldsymbol{x}_i \\ y_i \end{bmatrix}\right)$

$$\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$$

Step 1. Introduce $\boldsymbol{z} \in \{0,1\}^N$, use epigraphic reformulation trick

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} \boldsymbol{x}_i \\ y_i \end{bmatrix}\right)$

$$\min_{\substack{\boldsymbol{\theta} \\ \boldsymbol{z} \in \{0,1\}^N}} \frac{1}{N} \sum_{i=1}^{N} z_i$$

$$\text{s.t.} \quad z_i \geq \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0] \quad \forall i \in [N]$$

Binary Classification

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Step 2. Reformulate $z_i \geq \mathbb{I}[y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \leq 0]$, using Big-M trick

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$

$$\min_{\substack{\theta \\ z \in \{0,1\}^N}} \frac{1}{N} \sum_{i=1}^{N} z_i$$

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Step 2. Reformulate $z_i \geq \mathbb{I}[y_i(\theta^\top x_i) \leq 0]$, using Big-M trick

$$\min_{\substack{\theta \\ z \in \{0,1\}^N}} \frac{1}{N} \sum_{i=1}^{N} z_i$$

$$\text{s.t.} \quad M z_i \geq -y_i(\theta^\top x_i) \quad \forall i \in [N]$$

Binary Classification

Recall, with N samples $= \{\xi_1, \xi_2, \dots, \xi_N\}$ where $\left(\xi_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix}\right)$

$$\min_{\substack{\theta \\ z \in \{0,1\}^N}} \frac{1}{N} \sum_{i=1}^{N} z_i$$

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MILP problem since $(\theta, z) \in \mathbb{R}^{d+1} \times \{0,1\}^N$

Binary Classification

$$\min_{\substack{\boldsymbol{\theta} \\ \boldsymbol{z}\in\{0,1\}^N}} \frac{1}{N}\sum_{i=1}^{N} z_i$$

$$\text{s.t. } M z_i \geq -y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \quad \forall i \in [N]$$

Q1. Can you rewrite the problem in a canonical MILP?

$$\min_{\boldsymbol{x},\boldsymbol{z}} \boldsymbol{c}^\top \boldsymbol{x} + \boldsymbol{h}^\top \boldsymbol{z}$$

$$\text{s.t. } \boldsymbol{A}\boldsymbol{x} + \boldsymbol{G}\boldsymbol{z} \geq \boldsymbol{b}$$

$$\boldsymbol{x} \geq \boldsymbol{0}$$

$$\boldsymbol{z} \in \{0,1\}^l$$

Binary Classification

$$\min_{\substack{\boldsymbol{\theta} \\ \boldsymbol{z}\in\{0,1\}^N}} \frac{1}{N}\sum_{i=1}^{N} z_i$$

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$$\boldsymbol{x} \geq \boldsymbol{0}$$

$$\boldsymbol{z} \in \{0,1\}^l$$

$$\boldsymbol{c} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \boldsymbol{h} = \frac{1}{N}\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \boldsymbol{G} = M\begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}, \ldots$$

Binary Classification

$$\min_{\substack{\boldsymbol{\theta} \\ \boldsymbol{z}\in\{0,1\}^N}} \frac{1}{N}\sum_{i=1}^{N} z_i$$

$$\text{s.t. } Mz_i \geq -y_i(\boldsymbol{\theta}^\top \boldsymbol{x}_i) \quad \forall i \in [N]$$

Q2. Note that $dim(\boldsymbol{z}) = N$,

Solving the MILP for a very large N isn't good idea!

Can we approximate as a convex problem?

Binary Classification

$$\min_{\substack{\boldsymbol{\theta} \\ \boldsymbol{z}\in\{0,1\}^N}} \frac{1}{N}\sum_{i=1}^{N} z_i$$

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Solving the MILP for a very large N isn't good idea!

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Thanks!
Any question?

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